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Graphene-Integrated Metasurface for Full 2π Mid-Infrared Phase Modulation via GMR–Plasmon Strong Coupling

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***Abstract* – 2π phase modulation with uniform reflectance remains a key challenge in tunable metasurface design. We propose and numerically demonstrate a mid-infrared metasurface that achieves continuous 2π phase modulation at $\lambda = 7.41\ \mu\mathrm{m}$ under TM polarization by engineering a coupling between an overcoupled guided-mode resonance (GMR) and a graphene plasmonic (GP) mode. Electrically tuning the graphene Fermi level shifts the GP resonance across the fixed GMR frequency, inducing strong mode hybridization and phase repulsion while preserving uniform reflectance throughout the tuning range. The overcoupled GMR design ensures that hybridization redistributes the phase rather than suppressing amplitude, resulting in a near-circular trajectory of the reflection coefficient in the complex plane. The moderate-Q nature of the GMR reduces inter-pixel cross-talk, and combined with an inverse-design optimization framework enables effective beam steering and arbitrary wavefront shaping in the mid-infrared.**

I. Introduction

Electrically tunable phase modulation across the full 2π range with independently addressable pixels is a fundamental goal in modern photonics, as it enables complete control over optical wavefronts. Such local phase tunability is essential for applications, including beam steering and LiDAR systems [1, 2], holographic displays, wavefront shaping, and polarization manipulation. Beyond static operation, platforms capable of continuous and reversible 2π phase control are particularly promising for spatiotemporal modulation architectures that support dynamic beam steering and single-sideband carrier modulation.

Despite rapid advances in dynamic metasurfaces, two major constraints remain. First, modifying the resonance of a meta-atom generally perturbs both the phase and amplitude of the reflected or transmitted field simultaneously. This intrinsic phase–amplitude coupling often produces highly non-uniform intensity variations across the tuning range. The problem is further compounded by the considerable optical loss inherent to many tunable materials, particularly near resonance, where enhanced absorption can severely suppress the optical response. As a result, achieving efficient phase modulation without sacrificing reflectance remains a significant challenge. Second, the refractive-index modulation offered by most active materials is inherently limited. This restricted index variation constrains the achievable resonance shift, making it difficult to span the full 0–2π phase interval. Consequently, continuous and complete phase coverage with simultaneously high and nearly uniform optical amplitude has remained elusive in practical tunable metasurface platforms.

Controlling multi-parameter, like multi-gated structures, have demonstrated that full 2π phase shifts are attainable [2]. However, the introduction of multiple independent electrical control channels inevitably increases device complexity and reduces scalability. Moreover, previously reported implementations often exhibit low reflection amplitude and pronounced non-uniformity across the accessible phase range, limiting their suitability for practical wavefront engineering applications.

A recent approach based on the avoided crossing between a quasi-bound state in the continuum (qBIC) resonance and a graphene plasmonic (GP) mode has shown that full 2π phase modulation is achievable within a single-gate electrically tunable platform [3]. While this represents a significant advance, the strong nonlocal character of the qBIC resonance poses a fundamental barrier to independent pixel-level phase control. In this work, we design a metasurface with a moderate-Q factor GMR. Both guided-mode resonances (GMRs) and qBIC resonances

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are inherently nonlocal, arising from spatially extended, lattice-supported collective modes [4, 5]. However, the degree of nonlocality scales with the resonance quality factor: higher-Q modes exhibit longer photon lifetimes and greater spatial coherence lengths, resulting in stronger inter-pixel cross-talk [4]. Since qBIC resonances are typically engineered to achieve ultra-high Q-factors, their associated nonlocality have more severe barrier to independent per-pixel phase control compared to a moderate-Q, overcoupled GMR.

In this work, we overcome these limitations by exploiting the coupling between an overcoupled guided-mode resonance (GMR) and a graphene plasmonic (GP) mode under TM polarization at $\lambda = 7.41\,\mu$m. Electrical tuning of the graphene Fermi level shifts the GP resonance across the fixed GMR frequency, inducing strong mode hybridization and repulsion. This engineered avoided-crossing mechanism enables near-unity reflectance with a full 2π phase modulation range. The moderate-Q, overcoupled nature of the GMR results in weaker nonlocality compared to high-Q qBIC resonances, providing a more favorable starting point for wavefront engineering. We further demonstrate that the residual nonlocality can be effectively compensated through an inverse-design optimization framework, in which the geometric parameters of each unit cell are tailored to mitigate inter-cell coupling effects. This approach yields an improvement in beam steering efficiency compared to a naive forward-design strategy, providing a physically rigorous and electrically scalable pathway toward programmable mid-infrared metasurfaces.

II. Results and Discussion

A. Structure and Modeling

The metasurface consists of a periodic silicon bar array above a dielectric insulator backed by a gold reflector, as illustrated in Fig. 1a, operating in reflection mode. The geometric parameters of the structure are as follows: period $\Lambda = 2709$ nm, graphene strip width $w_g = 38.4$ nm, optical buffer thickness $h_{ins} = 163$ nm, guiding core thickness $h_{core} = 2173$ nm, insulator (HfO_2) thickness $h_{HfO_2} = 30$ nm, silicon bar thickness $h_{\text{bar}} = 3352$ nm, and silicon bar width $w_{\text{bar}} = 1613$ nm.

Under TM excitation (p-polarization), the structure supports a guided-mode resonance (GMR) with strong electric-field confinement near the top interface of the silicon , as shown in Fig. 1c. A monolayer graphene strip is placed at this electric-field maximum to ensure optimal modal overlap. The graphene supports a single TM plasmonic mode, modeled using the Kubo formalism with a carrier mobility of $\mu_s = 4000$ cm^2/V·s. The GMR is engineered in the overcoupled regime to preserve high reflectance during mode hybridization. Numerical simulations were performed using an in-house RCWA code with 400 Fourier orders, and a genetic algorithm was employed to optimize the structural parameters.

B. Strong Coupling Mechanism

The hybrid system can be described using a two-mode coupled Hamiltonian [3]:

$$H = \begin{bmatrix} \omega_{\text{GMR}} & u \\ u & \omega_{\text{GP}}(E_F) \end{bmatrix} + i \begin{bmatrix} \gamma_{\text{GMR},d} + \gamma_{\text{GMR},r} & \sqrt{\gamma_{\text{GMR},r}\gamma_{\text{GP},r}} \\ \sqrt{\gamma_{\text{GMR},r}\gamma_{\text{GP},r}} & \gamma_{\text{GP},d} + \gamma_{\text{GP},r} \end{bmatrix}, \tag{1}$$

where ω_{GMR} and $\omega_{\text{GP}}(E_F)$ are the uncoupled resonance frequencies of the GMR and graphene plasmon (GP), respectively, u is the near-field coupling strength, and γ_d and γ_r denote the dissipative and radiative decay rates of each mode. This formulation is adopted here to illustrate the avoided-crossing mechanism conceptually. The eigenvalues of H exhibit frequency repulsion as $\omega_{\text{GP}}(E_F)$ is tuned toward ω_{GMR}, producing the hybridized branches observed in Fig. 1g.

C. Amplitude and Phase Response

Figures 1f and 1g show the reflection phase and amplitude as functions of wavelength and Fermi level, respectively. A clear avoided crossing is observed near $\lambda \approx 7.41\,\mu$m and $E_F \approx 0.4$ eV, where the GMR and GP resonances repel each other and form two distinct hybridized branches. As the Fermi level is tuned across the coupling region, the phase map reveals a continuous 2π evolution while the reflection amplitude remains above 0.28 throughout the tuning range, confirming near phase-only modulation. This behavior is a direct consequence of the overcoupled GMR design, in which radiative decay dominates over intrinsic loss, so that hybridization primarily redistributes phase without significantly suppressing amplitude. As shown in Fig. 1e, this is further confirmed

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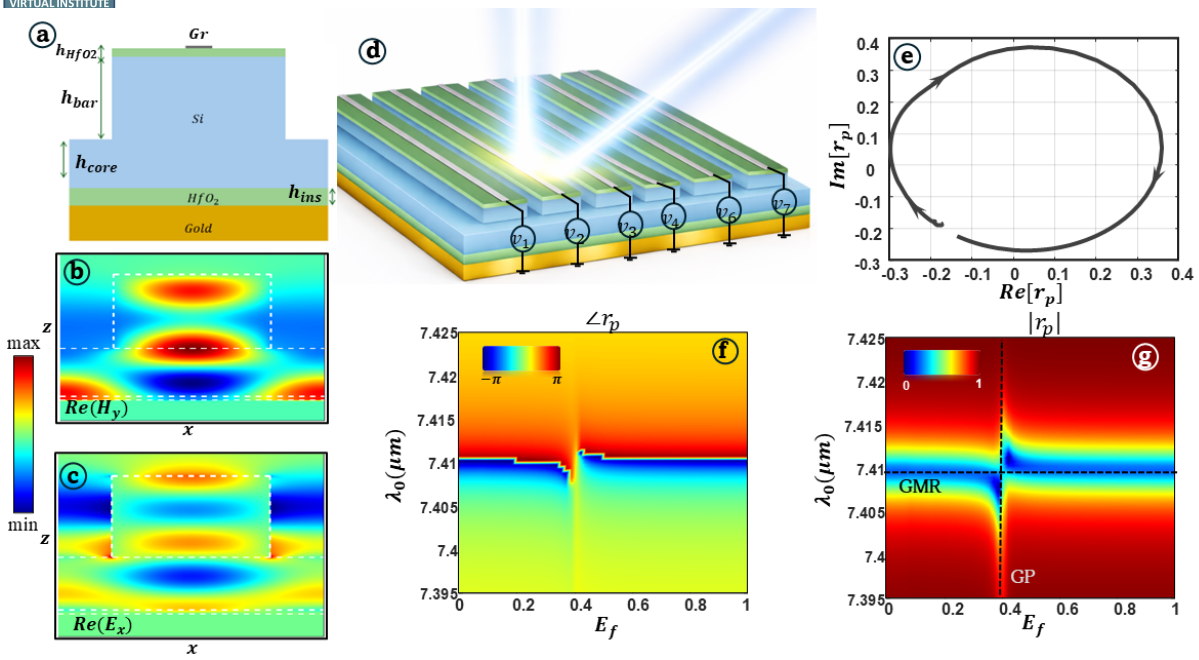


Fig. 1: (a) Schematic cross-section of the proposed metasurface unit-cell. (b,c) Simulated field distributions of $\mathrm{Re}(H_y)$ and $\mathrm{Re}(E_x)$ at the GMR resonance wavelength under TM excitation. (d) Schematic illustration of a periodic array of independently addressable unit cells proposed for beam steering, where each unit-cell is biased by a individually chosen gate voltage corresponding to a target Fermi level, enabling local phase control across the aperture. (e) Trajectory of the reflection coefficient r_p in the complex plane as the Fermi level is tuned, tracing a near-circular path indicative of near phase-only modulation with high reflectance. (f) Reflection phase $\angle r_p$ as a function of free-space wavelength λ_0 and Fermi level E_f (g) Reflection amplitude $|r_p|$ map showing the avoided crossing between the GMR and GP branches near $\lambda \approx 7.41\,\mu$m and $E_f \approx 0.4$ eV.

by the near-circular trajectory traced by the reflection coefficient in the complex plane. For graphene tuning, the silicon bar on top can be doped and used as an electrostatic gate to modulate the graphene Fermi level.

III. Conclusion

We have demonstrated that an engineered coupling between an overcoupled GMR and a graphene GP mode enable continuous 2π phase modulation while the reflection coefficient remains above 0.28 at $\lambda = 7.41\,\mu$m. The moderate-Q GMR design ensures near phase-only modulation and independent per-unit-cell electrical control, making this platform a promising candidate for inverse-design-based mid-infrared wavefront engineering, dynamic beam steering, and spatiotemporal modulation architectures. Additional studies are underway and will be presented on phase control and high-amplitude in the reflection mode.

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